

Sample MCQ – C Programming

1. Who is the father of C language?

- a) Bjarne Stroustrup
- b) Dennis Ritchie
- c) James Gosling
- d) Ken Thompson

Answer: b) Dennis Ritchie

2. C language was developed in the year:

- a) 1970
- b) 1972
- c) 1980
- d) 1990

Answer: b) 1972

3. Which symbol is used for a preprocessor directive?

- a) \$
- b) #
- c) *
- d) &

Answer: b) #

4. What is the correct extension of a C program file?

- a) .c
- b) .cpp
- c) .java
- d) .py

Answer: a) .c

5. Which of the following is NOT a valid C variable name?

- a) _count
- b) 123var
- c) count_1
- d) totalMarks

Answer: b) 123var

6. Which of the following is NOT a basic data type in C?

- a) int
- b) float
- c) char
- d) bool

Answer: d) bool

7. The sizeof() operator returns the size of a variable in:

- a) Bits
- b) Bytes
- c) Kilobytes
- d) None of the above

Answer: b) Byte

8. What is the size of int data type in C (assuming 32-bit system)?

- a) 2 bytes
- b) 4 bytes
- c) 8 bytes
- d) Depends on the system

Answer: b) 4 bytes

9. Which of the following is a logical operator in C?

- a) &&
- b) ||
- c) !
- d) All of the above

Answer: d) All of the above

10. The modulus operator (%) can be used with:

- a) float and int
- b) int only
- c) float only
- d) None of the above

Answer: b) int only

11. The default value of a local variable in C is:

- a) 0
- b) Garbage value
- c) NULL
- d) None of the above

Answer: b) Garbage value

12. What will be the output of:

```
int x = 10;  
if (x = 5)  
    printf("Hello");  
else  
    printf("World");
```

- a) Hello
- b) World
- c) Compile-time error
- d) None of the above

Answer: a) Hello

13. The break statement is used to exit from:

- a) Loop
- b) Switch case
- c) Both a and b
- d) None of the above

Answer: c) Both a and b

14. The continue statement in a loop:

- a) Exits the loop
- b) Skips the remaining statements in the loop and moves to the next iteration
- c) Repeats the loop indefinitely
- d) None of the above

Answer: b) Skips the remaining statements in the loop and moves to the next iteration

15. How many times will the following loop execute?

```
for (int i = 0; i < 5; i++)  
{  
    printf("%d", i);  
}
```

- a) 4 times
- b) 5 times
- c) Infinite loop
- d) 6 times

Answer: b) 5 times

16. Which keyword is used to return a value from a function?

- a) send
- b) return
- c) break
- d) continue

Answer: b) return

17. Which of the following is a correct way to declare a pointer?

- a) int ptr;
- b) int ptr;
- c) ptr int;
- d) int ptr;

Answer: b) int *ptr;

18. How do you allocate memory dynamically in C?

- a) malloc()
- b) calloc()
- c) realloc()
- d) All of the above

Answer: d) All of the above

19. Which of the following is the correct syntax to define a structure?

- a) struct { int a; float b; } s;
- b) struct s { int a; float b; };
- c) Both a and b
- d) None of the above

Answer: c) Both a and b

20. How do you open a file in write mode?

- a) fopen("file.txt", "r");
- b) fopen("file.txt", "w");
- c) fopen("file.txt", "rw");
- d) fopen("file.txt", "a");

Answer: b) fopen("file.txt", "w");

21. Which function is used to read a single character from a file?

- a) fgetc()
- b) fputc()
- c) fgets()
- d) fprintf()

Answer: a) fgetc()

22. Which library is required for file handling in C?

- a) stdio.h
- b) stdlib.h
- c) string.h
- d) conio.h

Answer: a) stdio.h

23. Which header file is required for memory allocation functions in C?

- a) stdio.h
- b) string.h
- c) stdlib.h
- d) math.h

Answer: c) stdlib.h

24. What does the realloc() function do?

- a) Allocates new memory
- b) Frees existing memory
- c) Resizes previously allocated memory
- d) None of the above

Answer: c) Resizes previously allocated memory

25. How do you deallocate memory allocated using malloc()?

- a) free()
- b) delete
- c) deallocate()
- d) None of the above

Answer: a) free()

26. Which of the following is NOT a valid loop in C?

- a) for
- b) while
- c) repeat
- d) do-while

Answer: c) repeat

27. What is the default return type of main() in C?

- a) void
- b) int
- c) char
- d) float

Answer: b) int

28. What is the correct way to declare a function that does not return any value?

- a) void functionName();
- b) int functionName();
- c) functionName();
- d) return functionName();

Answer: a) void functionName();

29. Which of the following storage classes provides the longest lifetime for a variable?

- a) auto
- b) static
- c) register
- d) extern

Answer: b) static

30. What does the #define preprocessor directive do?

- a) Defines a function
- b) Defines a constant or macro
- c) Allocates memory
- d) None of the above

Answer: b) Defines a constant or macro

31. What is the correct way to declare a function pointer?

- a) int *func();
- b) int (*func)();
- c) int func*();
- d) int func();

Answer: b) int (*func)();

32. What is the difference between struct and union?

- a) Both are identical
- b) struct stores all members, union shares memory
- c) union stores all members, struct shares memory
- d) None of the above

Answer: b) struct stores all members, union shares memory

33. Which of the following functions can be used to get the length of a string in C?

- a) length()
- b) strlen()
- c) sizeof()
- d) strlenlength()

Answer: b) strlen()

34. How do you define an array of 10 integers in C?

- a) int arr[10];
- b) array int arr[10];
- c) arr[10] int;
- d) int[10] arr;

Answer: a) int arr[10];

35. Which of the following is NOT a valid loop control statement in C?

- a) continue
- b) break
- c) exit
- d) jump

Answer: d) jump

36. What is the return type of malloc() in C?

- a) int
- b) void*
- c) float
- d) char*

Answer: b) void*

37. Which function is used to copy one string to another in C?

- a) strcpy()
- b) strncpy()
- c) memcpy()
- d) Both a and b

Answer: d) Both a and b

38. How do you declare a constant variable in C?

- a) constant int x = 10;
- b) const int x = 10;
- c) int const x = 10;
- d) Both b and c

Answer: d) Both b and c

39. What does `sizeof(int*)` return?

- a) Size of integer
- b) Size of pointer variable
- c) Size of address
- d) Both b and c

Answer: d) Both b and c (Typically 4 or 8 bytes depending on system)

40. What will happen if you use `free()` on a pointer that was not allocated using `malloc()`?

- a) Memory is freed
- b) Undefined behavior
- c) Compilation error
- d) No effect

Answer: b) Undefined behavior

41. What is the correct syntax for a function pointer?

- a) `int *func();`
- b) `int (*func)();`
- c) `int func*();`
- d) `int func();`

Answer: b) `int (*func)();`

42. What is the difference between `getchar()` and `scanf("%c", &ch);`?

- a) `getchar()` only reads a single character
- b) `scanf()` reads a character but ignores newline
- c) `getchar()` waits for input, `scanf()` does not
- d) Both a and b

Answer: d) Both a and b

43. Which library function is used to compare two strings?

- a) `strcmp()`
- b) `strcomp()`
- c) `strcompare()`
- d) `strcmp()`

Answer: a) `strcmp()`

44. What is the keyword used to prevent modification of a variable?

- a) `static`
- b) `constant`
- c) `const`
- d) `final`

Answer: c) `const`

45. What is the correct way to declare a pointer to a function returning an integer?

- a) `int *func();`
- b) `int (*func)();`
- c) `int func*();`
- d) `int func();`

Answer: b) `int (*func)();`

46. How can you determine the size of a variable in C?

- a) `sizeof()`
- b) `size()`
- c) `lengthof()`
- d) `memoryof()`

Answer: a) `sizeof()`

47. Which function is used to copy memory from one location to another?

- a) `memcpy()`
- b) `memcopy()`
- c) `copy()`
- d) `move()`

Answer: b) `memcopy()`

48. What does static keyword do for a local variable?

- a) Increases visibility
- b) Maintains value between function calls
- c) Allocates memory dynamically
- d) None of the above

Answer: b) Maintains value between function calls

49. What is the result of 5 & 3 in binary operation?

- a) 1
- b) 3
- c) 5
- d) 2

Answer: d) 2 (Bitwise AND operation)

50. Which of the following is not a valid storage class in C?

- a) `auto`
- b) `register`
- c) `mutable`
- d) `extern`

Answer: c) `mutabl`